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Dashboard Screenshot

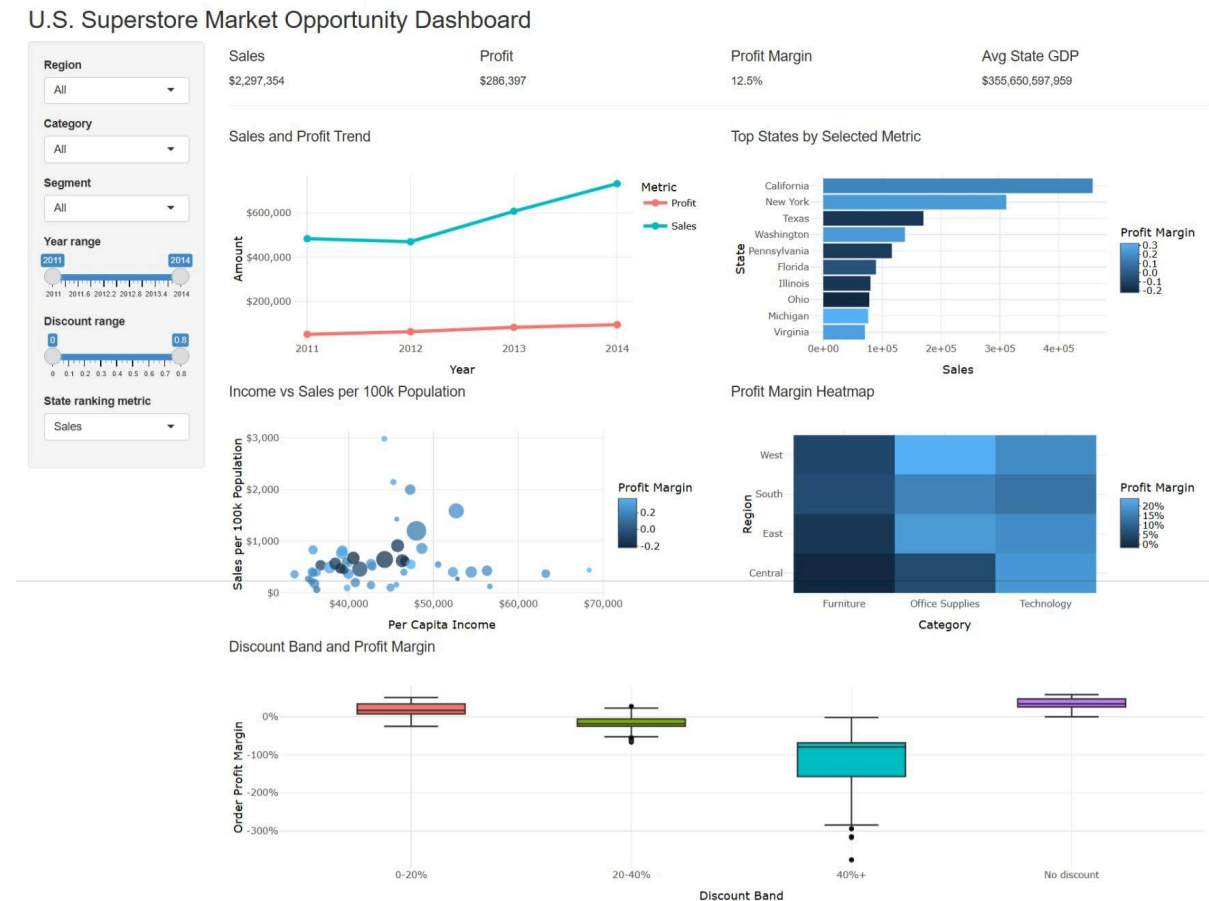


Figure 1: U.S. Superstore Market Opportunity Dashboard

Figure 1 shows the complete R Shiny dashboard developed for this assignment. The dashboard combines the Superstore dataset with three external state-level datasets: state income, state population and state GDP. The dashboard includes interactive filters for Region, Category, Segment, Year range, Discount range and State ranking metric. It also displays key performance indicators for Sales, Profit, Profit Margin and Average State GDP.

Based on the current dashboard view, total Sales are \$2,297,354, total Profit is \$286,397, Profit Margin is 12.5%, and Average State GDP is \$355,650,597,959. The dashboard is designed to help business users identify which U.S. states and market conditions offer stronger opportunities for profitable growth.

Problem Design

Problem Statement

The dashboard addresses the problem of identifying which U.S. Superstore markets should be prioritised for profitable growth. The Superstore dataset provides detailed information about sales, profit, discount, product category, customer segment, region and state. However, internal sales data alone may not provide a fair comparison across states because larger states or stronger economies may naturally generate higher sales.

Therefore, this dashboard combines Superstore data with external state-level income, population and GDP data. This allows business performance to be evaluated together with economic and demographic context. The main problem addressed is:

Which U.S. states, product categories and discount conditions offer the strongest opportunities for profitable growth when Superstore sales and profit are evaluated together with state-level income, population and GDP?

Dashboard Objectives

The dashboard has four main objectives. First, it provides an overview of sales and profit performance over time. Second, it identifies the top-performing states based on a selected performance metric. Third, it examines whether state income and population-adjusted sales are related. Fourth, it analyses how product categories and discount levels affect profit margin.

The cleaned datasets used in this dashboard were prepared by cleaning the Superstore dataset, reshaping the external state income, GDP and population datasets into long format, merging them by state and year, and exporting the final cleaned datasets for use in the R Shiny dashboard.

INFLUENCE Framework Application

INFLUENCE Element	Application in This Dashboard
Intent Alignment	The dashboard is designed for business decision-makers who need to decide where Superstore should focus profitable growth. The focus is on sales, profit margin, state opportunity, product category performance and discount impact.
Narrative Arc	The dashboard begins with overall KPIs and the sales/profit trend. It then identifies leading states, examines economic context through income and sales per 100,000 population, compares profitability by region and category, and ends by showing the effect of discounts on profit margin.
Framing for Cognitive Biases	The dashboard avoids relying only on total sales. It also includes profit margin, sales per 100,000 population and sales relative to GDP. This reduces the risk of assuming that the largest sales markets are automatically the best growth opportunities.
Logic Chain	State economic conditions and population affect market potential. Product category and discounting affect profitability. Therefore, profitable growth should be based on both market size and margin quality.
Urgency Creation	The dashboard shows that although total sales are high at \$2,297,354, total profit is much smaller at \$286,397, giving a profit margin of 12.5%. The discount box plot further shows that heavy discounting, especially 40%+, is linked to strongly negative profit margins.
Emotional Resonance	The dashboard connects the analysis to practical business concerns by showing that unprofitable growth can reduce business value. Heavy discounting may increase sales but damage profit, so each poor-margin transaction represents a missed opportunity to grow sustainably.
Next Action	Managers can use the dashboard to prioritise high-performing states, investigate weaker profit-margin markets, focus on stronger categories such as Technology, and review discount strategies.
Credibility	The dashboard combines internal Superstore data with external state income, population and GDP data. This improves the reliability of the analysis because state performance is interpreted with broader economic context.

Overall, the dashboard is designed as a business decision tool rather than only a set of charts. It helps users move from observing overall performance to identifying specific market opportunities and profitability risks.

Dataset Selection

Superstore Dataset

The Superstore dataset is the main internal dataset used in this dashboard. It contains order-level business information, including sales, profit, discount, quantity, category, segment, region, state and year. It provides the core business performance data for analysing Superstore's sales and profitability across markets.

External Dataset 1: State Income Dataset

Citation: U.S. Bureau of Economic Analysis. (n.d.). *SAINCI State annual personal income summary: Personal income, population, per capita personal income (2011–2014)* [Data set]. U.S. Department of Commerce.

Justification: This dataset was selected because per capita personal income reflects the purchasing power of each state. It helps assess whether higher-income states also show stronger Superstore sales intensity.

Description: The dataset used is `state_income.csv`. It contains state-level per capita personal income values for 2011 to 2014. The year columns were reshaped into long format and renamed as `Per_Capita_Income`.

External Dataset 2: State Population Dataset

Citation: U.S. Census Bureau. (2021, October 8). *Annual population estimates, estimated components of resident population change, and rates of the components of resident population change for the United States, states, the District of Columbia, and Puerto Rico: April 1, 2010 to July 1, 2020 (NST-EST2020-alldata)* [Data set]. U.S. Department of Commerce.

Justification: This dataset was selected because raw sales totals can be misleading when comparing states with different population sizes. Population data allows sales per 100,000 population to be calculated for fairer market comparison.

Description: The dataset used is `state_population.csv`. It contains annual state population estimates. The variables `POPESTIMATE2011`, `POPESTIMATE2012`, `POPESTIMATE2013` and `POPESTIMATE2014` were reshaped into long format and renamed as `Population`.

External Dataset 3: State GDP Dataset

Citation: U.S. Bureau of Economic Analysis. (n.d.). *SAGDPI State annual gross domestic product (GDP) summary (2011–2014)* [Data set]. U.S. Department of Commerce.

Justification: This dataset was selected because GDP reflects the economic size of each state. It helps compare Superstore performance with broader state economic activity.

Description: The dataset used is `state_gdp.csv`. It contains state-level GDP values for 2011 to 2014. The year columns were reshaped into long format and renamed as `GDP_Million`.

Dataset Integration

The Superstore dataset provides internal business performance data, while the external datasets add economic and demographic context. Income supports purchasing power analysis, population supports population-adjusted sales comparison, and GDP reflects state economic size. Together, these datasets allow the dashboard to evaluate market opportunity more fairly than using sales and profit alone.

Data Preparation Summary

Before dashboard development, the datasets were cleaned and standardised. The Superstore dataset was filtered to U.S. records, while the state income, population and GDP datasets were reshaped into long format for 2011–2014. All external datasets were then merged by State and Year, allowing Superstore performance to be analysed together with economic and demographic context.

Visualisation 1 : Sales and Profit Trend

Sales and Profit Trend

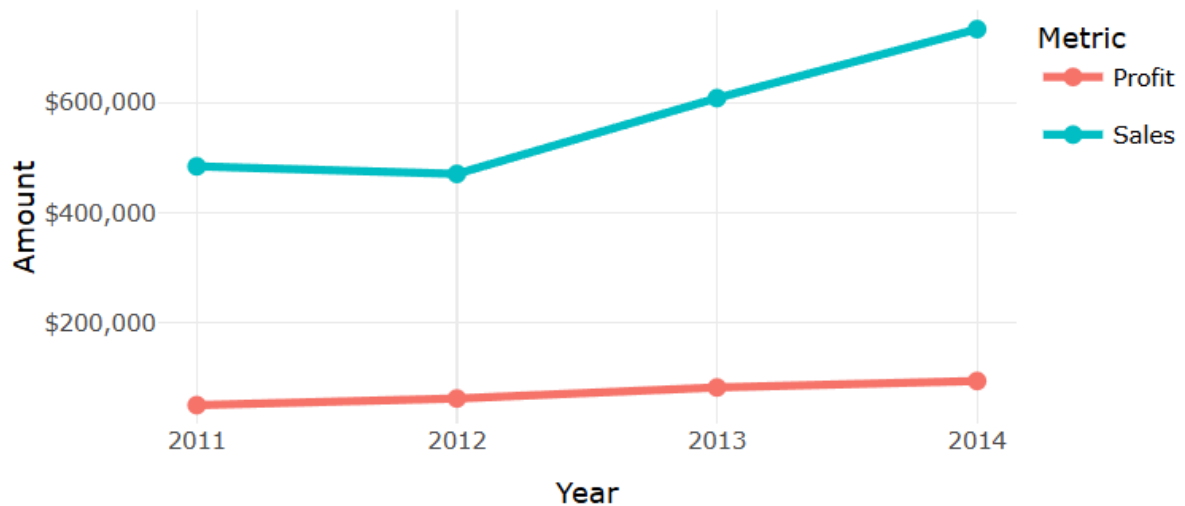


Figure 2: Sales and Profit Trend

Figure 2 shows Superstore sales and profit from 2011 to 2014. Sales remained relatively stable between 2011 and 2012, then increased strongly in 2013 and 2014. Profit also increased gradually across the period, but it remained much lower than sales in absolute value.

This chart contributes to the dashboard problem by showing the overall business direction before deeper market analysis. The upward sales trend suggests that Superstore experienced growth during the period. However, the large gap between sales and profit shows that revenue growth does not automatically translate into equally strong profitability. This supports the need to examine profit margin, category performance and discount behaviour in later visualisations.

Visualisation 2 : Top States by Selected Metric

Top States by Selected Metric

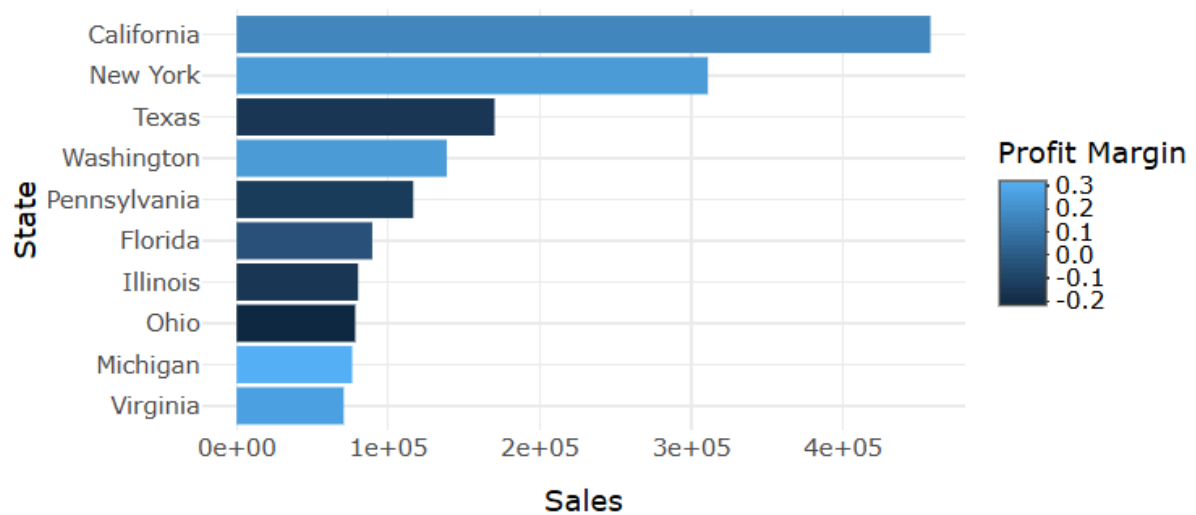


Figure 3: Top States by Selected Metric

Figure 3 ranks the top ten states by the selected metric. In the screenshot, the selected metric is Sales. California is the highest-performing state by sales, followed by New York and Texas. Washington and Pennsylvania are also among the stronger states, while Florida, Illinois, Ohio, Michigan and Virginia complete the top ten.

This chart directly supports the dashboard objective because it identifies which states contribute most strongly to Superstore performance. However, the chart also uses colour to show profit margin, which highlights that high sales do not always mean high profitability. This is important because a business should not prioritise markets based only on sales volume. A state with strong sales but weak profit margin may require pricing, discount or product mix review before further expansion.

Visualisation 3 : Income vs Sales per 100k Population

Income vs Sales per 100k Population

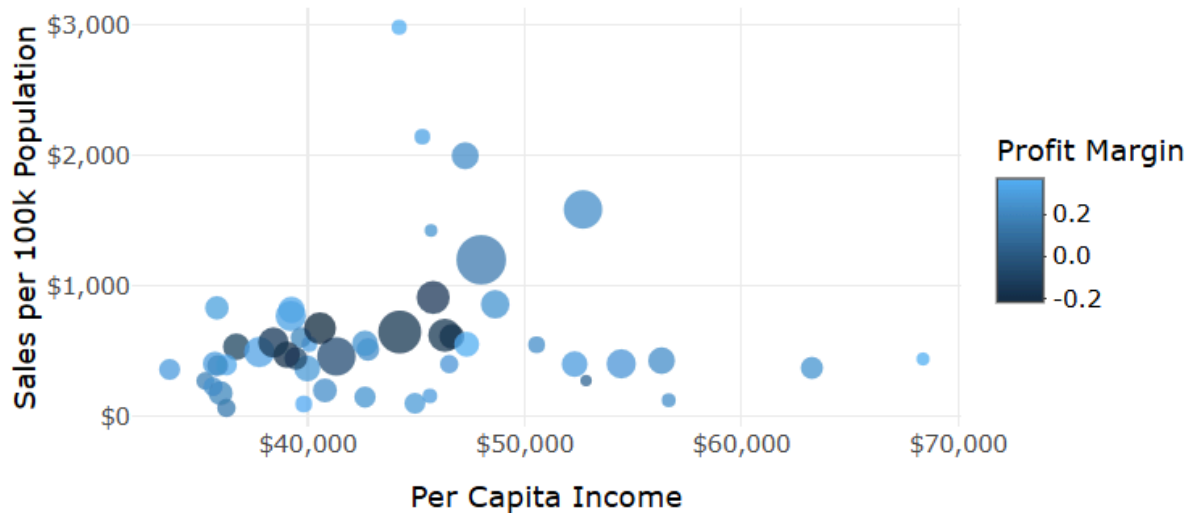


Figure 4: Income vs Sales per 100k Population

Figure 4 compares state per capita income with Superstore sales per 100,000 population. Each point represents a state. The size of each point represents population, while the colour represents profit margin. Most states are clustered between approximately \$35,000 and \$50,000 in per capita income, with sales per 100,000 population generally below \$1,000. A few states appear as higher sales-intensity outliers.

This chart shows that income is a useful context, but it does not fully explain Superstore performance. Some states with moderate income levels appear to generate stronger sales per 100,000 population than some higher-income states. This suggests that market opportunity depends on more than income alone. Factors such as customer demand, product mix, regional strategy and business penetration may also affect sales intensity.

Visualisation 4 : Profit Margin Heatmap

Profit Margin Heatmap

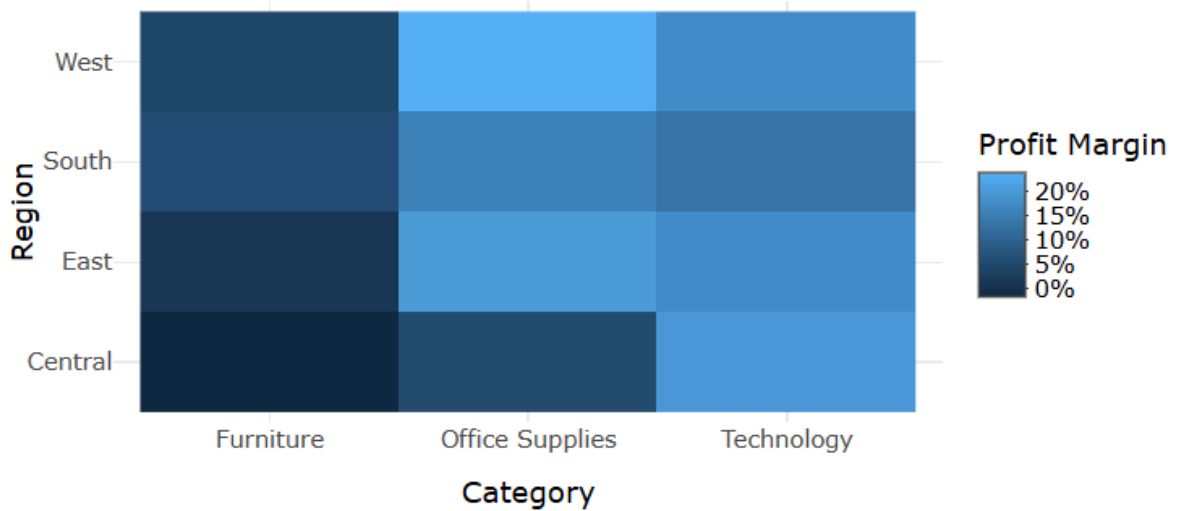


Figure 5: Profit Margin Heatmap

Figure 5 compares profit margin across regions and product categories. The heatmap shows that Technology generally has stronger profit margins across regions, while Furniture appears weaker. Office Supplies performs between these two categories, with stronger margins than Furniture in several regions.

This visualisation helps explain why some areas may be more profitable than others. A state or region may generate high sales, but if sales are concentrated in a weaker-margin category, the business may not achieve strong profit. The chart suggests that profitable growth should consider category mix, not only geographic market size. Technology appears to be an important category for margin strength, while Furniture requires closer attention.

Visualisation 5 : Discount Band and Profit Margin

Discount Band and Profit Margin

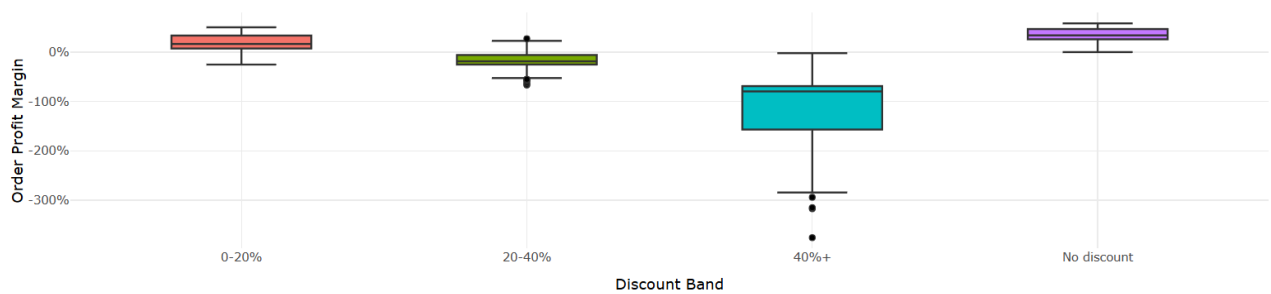


Figure 6: Discount Band and Profit Margin

Figure 6 shows the distribution of order profit margin across different discount bands. Orders with No discount have the strongest positive profit margins. Orders with 0–20% discounts are still generally profitable, but the margin is lower than no-discount orders. The 20–40% discount band shows weaker margins, with many values around or below zero. The 40%+ discount band performs the worst, with strongly negative margins and several severe negative outliers.

This chart is one of the most important visuals for explaining profitability. It shows that discounting has a strong effect on profit margin. While small discounts may still be manageable, heavy discounts appear to damage profitability significantly. Therefore, if Superstore wants profitable growth, it should not rely too heavily on aggressive discounting. Discount policy should be reviewed especially in markets or categories where sales are high but profit margin is weak.

Data Story

What is happening?

Superstore is generating strong overall business activity. The dashboard shows total sales of \$2,297,354 and total profit of \$286,397, resulting in a profit margin of 12.5%. These figures suggest that the business is profitable overall, but the difference between sales and profit also shows that margin control is important.

The key issue is that not all sales are equally valuable. High sales can come from large states, strong economic conditions, popular product categories or heavy discounting. Therefore, the business needs to understand which markets are truly profitable and which sales may be less valuable because of weaker margins.

How is performance changing over time?

The sales and profit trend shows that Superstore performance improved between 2011 and 2014. Sales increased strongly after 2012, especially in 2013 and 2014. Profit also increased during the period, showing that the business was able to generate more earnings over time.

However, profit increased at a much smaller scale than sales. This means that the company should not focus only on increasing revenue. It should also investigate whether discounts, category mix or market conditions are limiting profitability.

Which states are leading performance?

The state ranking chart shows that California is the leading state by sales, followed by New York and Texas. These states are important because they contribute a large share of total sales. Washington and Pennsylvania also appear among the stronger states.

However, the chart also shows profit margin through colour. This indicates that state performance should not be judged only by sales amount. Some high-sales states may have weaker margins, while some smaller states may perform more efficiently. Therefore, state prioritisation should consider both sales volume and profitability.

What role do economic conditions play?

The income versus sales per 100,000 population chart shows that higher income does not automatically lead to higher Superstore sales intensity. Many states are clustered around similar income levels, but their sales per 100,000 population differ. This means income is useful for understanding market context, but it is not the only driver of performance.

Some states with moderate income levels appear to perform relatively well, while some higher-income states do not stand out strongly. This suggests that Superstore should look for underdeveloped opportunities in states where income or GDP is strong but sales intensity is not yet high.

What is driving profitability?

The profit margin heatmap shows that profitability differs by category and region. Technology appears to be the strongest category in terms of margin, while Furniture is weaker. This suggests that product mix is an important factor in profitable growth.

The discount box plot gives an even clearer explanation of profit margin differences. No-discount orders generate the strongest margins, while heavy discounts are associated with severe margin losses. The 40%+ discount band is especially risky because it produces strongly negative profit margins.

What should the business do next?

Based on the dashboard, Superstore should focus on three actions. First, it should continue monitoring high-sales states such as California, New York and Texas, but should evaluate them using profit margin and population-adjusted measures, not only total sales. Second, it should strengthen higher-margin categories such as Technology and review weaker-margin areas such as Furniture. Third, it should reduce reliance on heavy discounting, especially discounts above 40%, because these are strongly associated with negative profit margins.

Overall conclusion

The dashboard shows that profitable growth is not simply about selling more. It depends on selecting the right states, understanding economic context, focusing on profitable categories and controlling discount levels. By combining Superstore data with state income, population and GDP data, the dashboard provides a more complete basis for identifying market opportunities and improving profitability.

